

METODE PEMULUSAN WINTER ADITIF & WINTER MULTIPLIKATIF

Pertemuan ke-4
Akbar Rizki, M.Si

OUTLINE

1. Pemulusan untuk Data Musiman

2. Pemulusan Winter Aditif

3. Pemulusan Winter Multiplikatif

OUTLINE

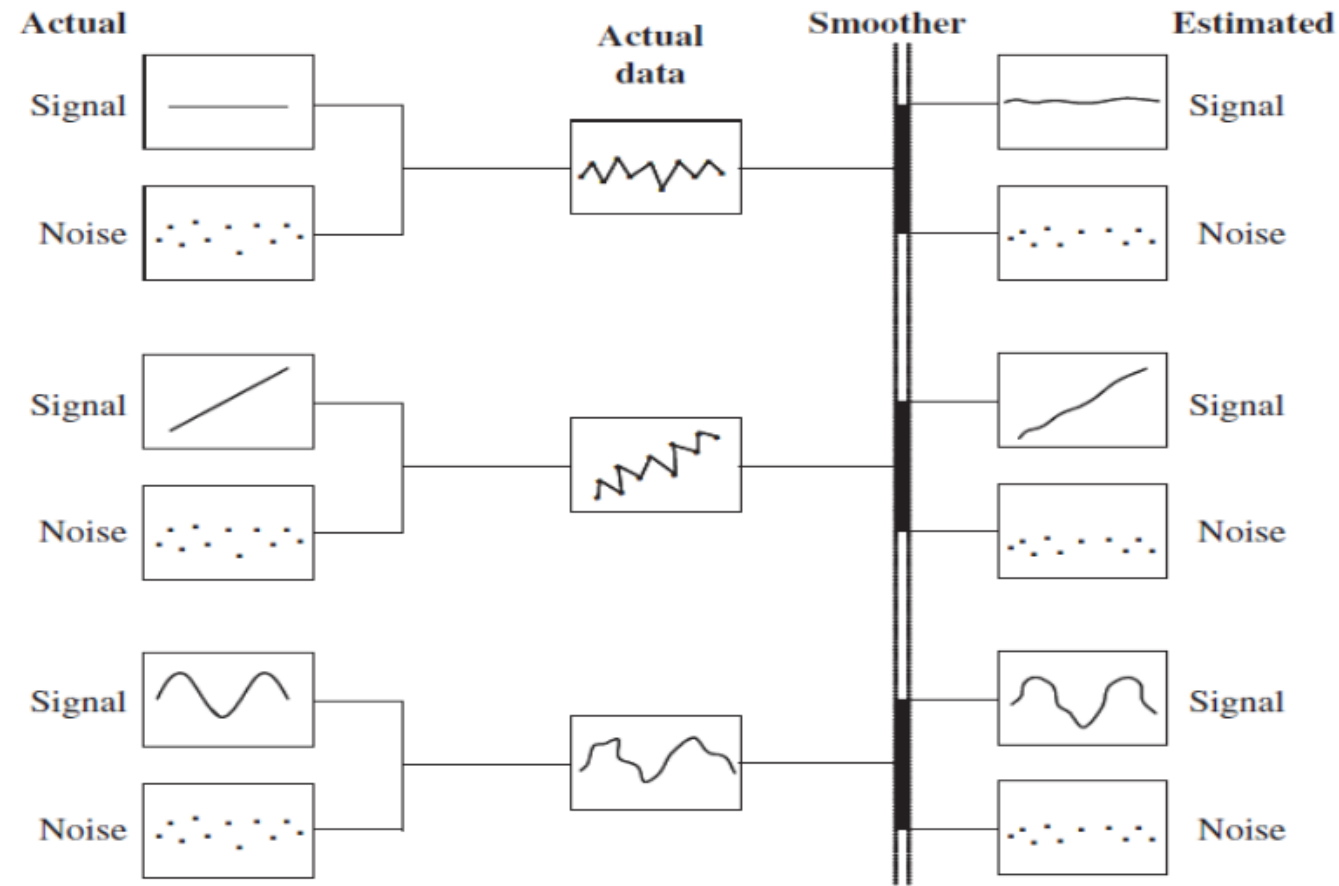
1. Pemulusan untuk Data Musiman

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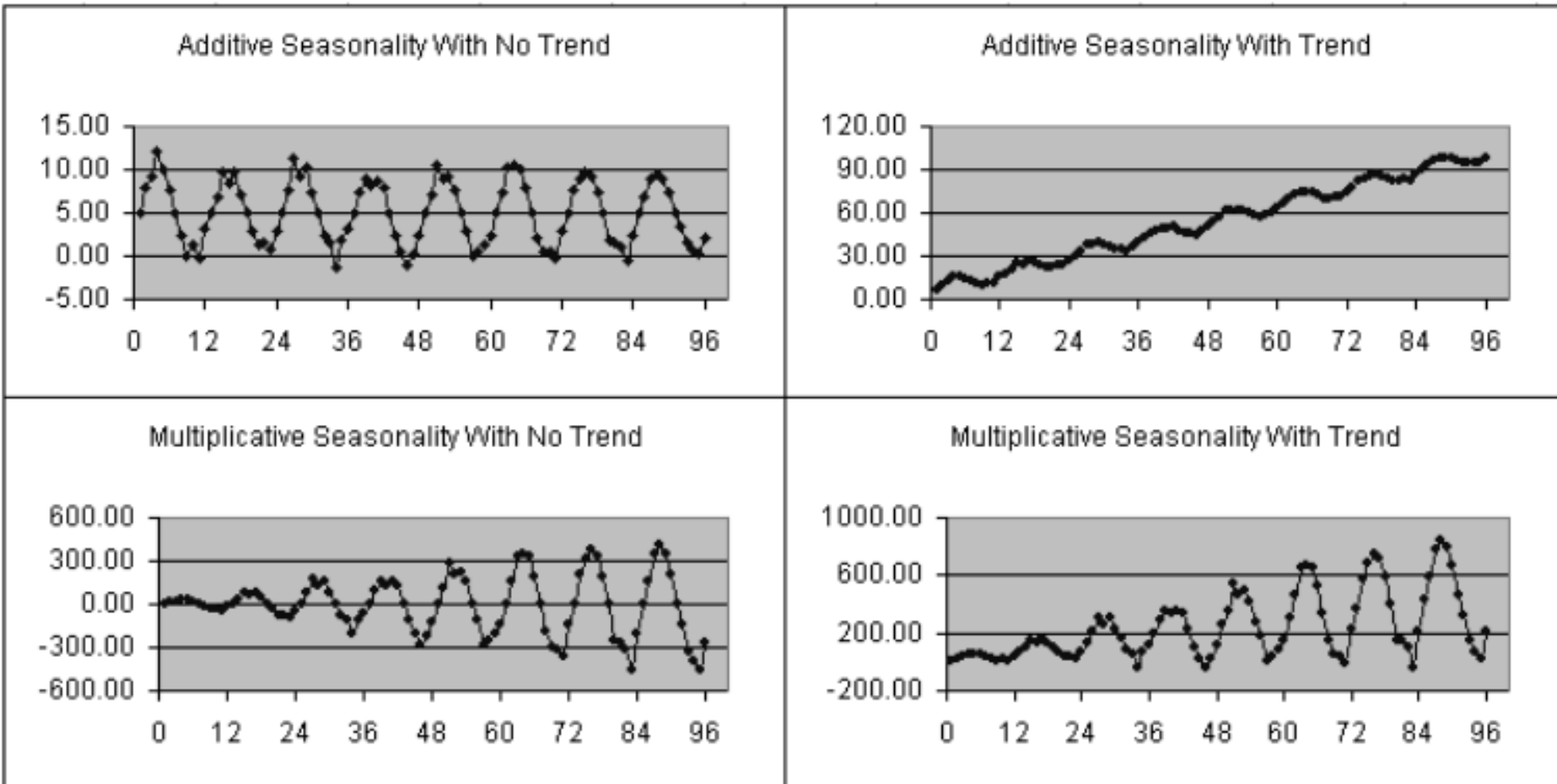
1. PEMULUSAN UNTUK DATA MUSIMAN

REVIEW PEMULUSAN



1. PEMULUSAN UNTUK DATA MUSIMAN

DATA MUSIMAN



1. PEMULUSAN UNTUK DATA MUSIMAN

ILUSTRASI

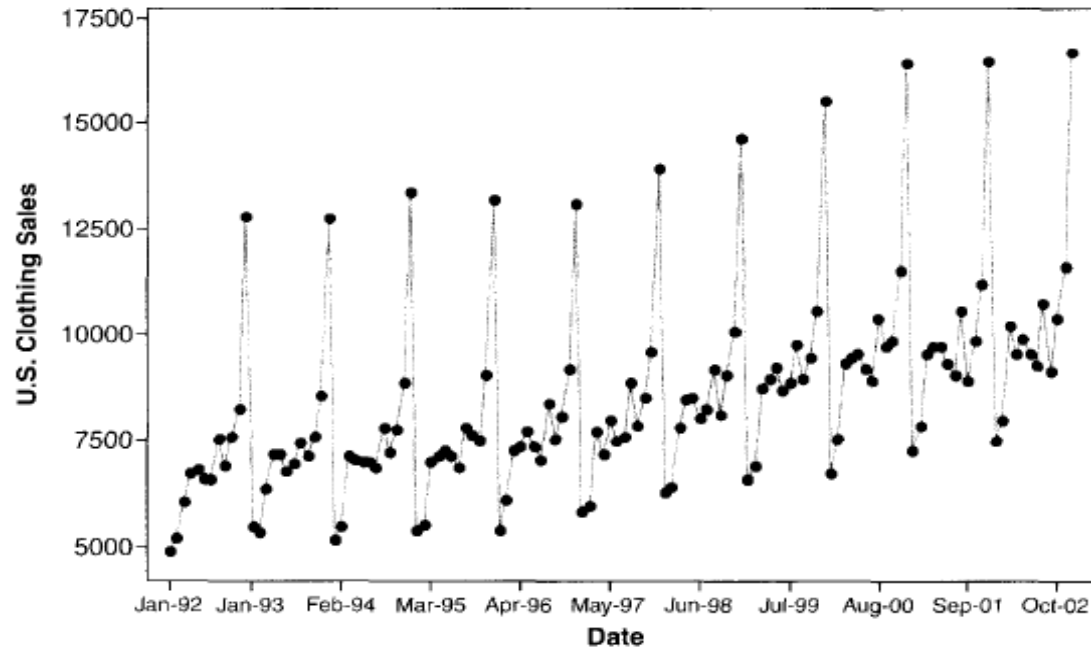


FIGURE 4.26 Time series plot of U.S. clothing sales from January 1992 to December 2003.

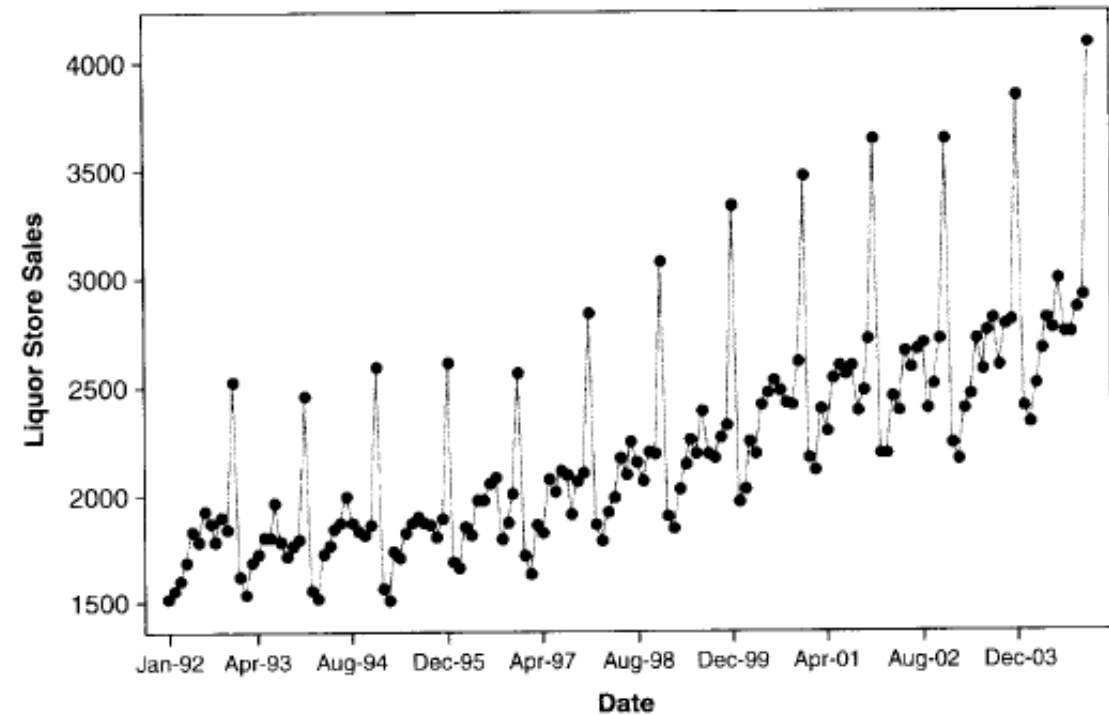


FIGURE 4.29 Time series plot of liquor store sales data from January 1992 to December 2004.

1. PEMULUSAN UNTUK DATA MUSIMAN

EXPONENTIAL SMOOTHING FOR SEASONAL DATA

- Originally introduced by Holt (1957) and Winters (1960)
- Generally known as Winters' method
- Basic idea: seasonal adjustment -> linear trend model
- Two types of adjustments are suggested:
 1. Additive
 2. Multiplicative

OUTLINE

1. Pemulusan untuk Data Musiman

2. Pemulusan Winter Aditif

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2. PEMULUSAN WINTER ADITIF

Additive Model

$$y_t = L_t + S_t + \varepsilon_t$$

level or linear trend component
can in turn be represented by $\beta_0 + \beta_1 t$

the seasonal adjustment

$$S_t = S_{t+m} = S_{t+2m} = \dots \text{ for } t = 1, \dots, m-1$$

length of the season (period) of the cycles

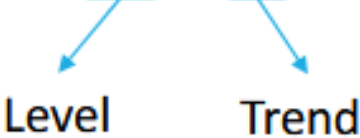
$$\sum_{t=1}^m S_t = 0.$$

2. PEMULUSAN WINTER ADITIF

Double Exponential Vs Additive Holt-Winter's Method

Double Exponential: $\hat{y}_{t+h}(t) = L_t + T_t h$

Level Trend



Holt-Winter: $\hat{y}_{t+h}(t) = L_t + B_t h + S_{t+h-m}$

Level Trend Seasonal



Holt Winter $\approx$ Triple Exponential Smoothing



2. PEMULUSAN WINTER ADITIF

Holt-Winters Additive Formulation

- Suppose the time series is denoted by $y_1, \dots, y_n$ with m seasonal period.

Estimate of the level:

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

Estimate of the trend:

$$b_t = \gamma(l_t - l_{t-1}) + (1 - \gamma)b_{t-1}$$

Estimate of the seasonal factor:

$$s_t = \delta(y_t - l_t) + (1 - \delta)s_{t-m}$$

- Let $\hat{y}_{t+h}(t)$ be the h -step forecast made using data to time t

$$\hat{y}_{t+h}(t) = l_t + b_t h + s_{t+h-m}$$

2. PEMULUSAN WINTER ADITIF

The Procedure

Step 1: Initialize the value of l_t , b_t , and s_t

Step 2: Update the estimate of l_t

Step 3: Update the estimate of b_t

Step 4: Update the estimate of s_t

Step 5: Conduct the h -step-ahead forecast

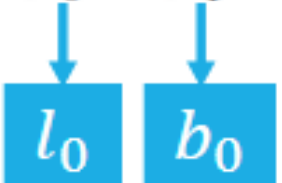
2. PEMULUSAN WINTER ADITIF

Initializing the Holt-Winters method

Montgomery (2015):

use the least squares estimates of the following model:

$$y_t = \beta_0 + \beta_1 t + \sum_{i=1}^{s-1} \gamma_i (I_{t,i} - I_{t,s}) + \varepsilon_t,$$



where

$$I_{t,i} = \begin{cases} 1, & t = i, i + s, i + 2s, \dots \\ 0, & \text{otherwise} \end{cases}.$$

$$\hat{s}_{j-s} = \hat{y}_j \text{ for } 1 \leq j \leq m - 1, \text{ and } \hat{s}_0 = -\sum_{j=1}^{m-1} \hat{y}_j$$

2. PEMULUSAN WINTER ADITIF

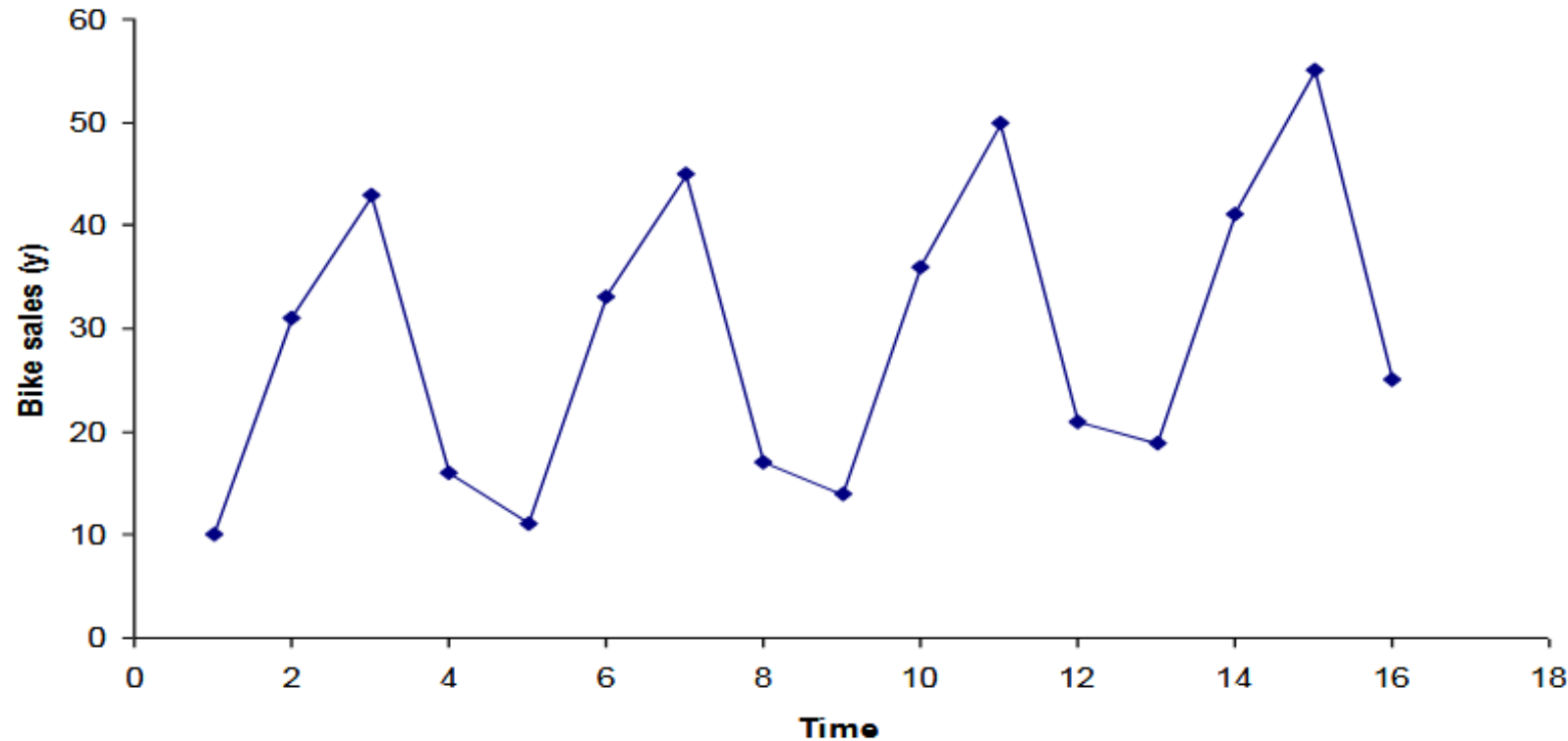
Procedures of Additive Holt-Winters Method

Consider the Mountain Bike example,

Quarterly sales of the TRK-50 Mountain Bike				
	Year			
Quarter	1	2	3	4
1	10	11	14	19
2	31	33	36	41
3	43	45	50	55
4	16	17	21	25

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method



Observations:

- Linear upward trend over the 4-year period
 - Magnitude of seasonal span is almost constant as the level of the time series increases
- *Additive Holt-Winters method can be applied to forecast future sales*

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method

Step 1: Obtain initial values for the level ℓ_0 , the growth rate b_0 , and the seasonal factors sn_{-3} , sn_{-2} , sn_{-1} , and sn_0 , by fitting a least squares trend line to at least four or five years of the historical data.

- y -intercept = ℓ_0 ; slope = b_0

Example

- Fit a least squares trend line to all 16 observations
- Trend line

$$\hat{y}_t = 20.85 + 0.980882t$$
$$\ell_0 = 20.85; b_0 = 0.9809$$

SUMMARY OUTPUT	
<i>Regression Statistics</i>	
Multiple R	0.320508842
R Square	0.102725918
Adjusted R Square	0.038634912
Standard Error	14.28614022
Observations	16
ANOVA	
	<i>df</i>
Regression	1
Residual	14
Total	15
<i>Coefficients</i>	
Intercept	20.85
Time	0.980882353

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method

Step 2: Find the initial seasonal factors

1. Compute $\hat{y}_t$ for each time period that is used in finding the least squares regression equation. In this example, $t = 1, 2, \dots, 16$.

$$\hat{y}_1 = 20.85 + 0.980882(1) = 21.8309$$

$$\hat{y}_2 = 20.85 + 0.980882(2) = 22.8118$$

.....

$$\hat{y}_{16} = 20.85 + 0.980882(16) = 36.5441$$

Step 2: Find the initial seasonal factors

3. Compute the average seasonal values for each of the L seasons. The L averages are found by computing the average of the detrended values for the corresponding season. For example, for quarter 1,

$$\begin{aligned}\bar{S}_{[1]} &= \frac{S_1 + S_5 + S_9 + S_{13}}{4} \\ &= \frac{(-11.8309) + (-14.7544) + (-15.6779) + (-14.6015)}{4} = -14.2162\end{aligned}$$

Step 2: Find the initial seasonal factors

2. Detrend the data by computing $S_t = y_t - \hat{y}_t$ for each observation used in the least squares fit. In this example, $t = 1, 2, \dots, 16$.

$$S_1 = y_1 - \hat{y}_1 = 10 - 21.8309 = -11.8309$$

$$S_2 = y_2 - \hat{y}_2 = 31 - 22.8112 = 8.1882$$

.....

$$S_{16} = y_{16} - \hat{y}_{16} = 25 - 36.5441 = -11.5441$$

Step 2: Find the initial seasonal factors

4. Compute the average of the L seasonal factors. The average should be 0.

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method

Step 3: Calculate a point forecast of y_1 from time 0 using the initial values

$$\hat{y}_{T+p}(T) = \lambda_T + pb_T + sn_{T+p-L} \quad (T = 0, p = 1)$$

$$\begin{aligned}\hat{y}_1(0) &= \lambda_0 + b_0 + sn_{1-4} = \lambda_0 + b_0 + sn_{-3} \\ &= 20.85 + 0.9809 + (-14.2162) = 7.6147\end{aligned}$$

Step 4: Update the estimates ℓ_T , b_T , and sn_T by using some predetermined values of smoothing constants.

Example: let $\alpha = 0.2$, $\gamma = 0.1$, and $\delta = 0.1$

$$\begin{aligned}\lambda_1 &= \alpha(y_1 - sn_{1-4}) + (1 - \alpha)(\lambda_0 + b_0) \\ &= 0.2(10 - (-14.2162)) + 0.8(20.85 + 0.9808) = 22.3079\end{aligned}$$

$$\begin{aligned}b_1 &= \gamma(\lambda_1 - \lambda_0) + (1 - \gamma)b_0 \\ &= 0.1(22.3079 - 20.85) + 0.9(0.9809) = 1.0286\end{aligned}$$

$$\begin{aligned}sn_1 &= \delta(y_1 - \lambda_1) + (1 - \delta)sn_{1-4} \\ &= 0.1(10 - 22.3079) + 0.9(-14.2162) = -14.0254\end{aligned}$$

$$\begin{aligned}\hat{y}_2(1) &= \lambda_1 + b_1 + sn_{2-4} = \lambda_1 + b_1 + sn_{-2} \\ &= 22.3079 + 1.0286 + 6.5529 = 29.8895\end{aligned}$$

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method

1	n	alpha	gamma	delta	SSE	MSE	s	
2	16	0.2000	0.1000	0.1000	25.2166	1.9397	1.3927	
3								
4								
5						Forecast		Squared
6				Growth	Seasonal	Made Last	Forecast	Forecast
7	Time	y	Level	Rate	Factor	Period	Error	Error
8	-3				-14.2162			
9	-2				6.5529			
10	-1				18.5721			
11	0		20.85	0.9809	-10.9088			
12	1	10	22.30794	1.0286	-14.0254	7.6147	2.3853	5.6896
13	2	31	23.55864	1.0508	6.6418	29.8895	1.1105	1.2333
14	3	43	24.57314	1.0472	18.5575	43.1815	-0.1815	0.0329
15	4	16	25.87801	1.0729	-10.8057	14.7115	1.2885	1.6603
16	5	11	26.56583	1.0344	-14.1794	12.9256	-1.9256	3.7079
17	6	33	27.35185	1.0096	6.5424	34.2420	-1.2420	1.5427
18	7	45	27.97764	0.9712	18.4040	46.9190	-1.9190	3.6825
19	8	17	28.72023	0.9483	-10.8972	18.1431	-1.1431	1.3067
20	9	14	29.37074	0.9186	-14.2985	15.4892	-1.4892	2.2176
21	10	36	30.12295	0.9019	6.4759	36.8317	-0.8317	0.6918
22	11	50	31.1391	0.9133	18.4497	49.4289	0.5711	0.3262
23	12	21	32.0214	0.9102	-10.9096	21.1553	-0.1553	0.0241
24	13	19	33.00502	0.9176	-14.2692	18.6331	0.3669	0.1346
25	14	41	34.04291	0.9296	6.5240	40.3985	0.6015	0.3618
26	15	55	35.28807	0.9612	18.5759	53.4222	1.5778	2.4894
27	16	25	36.18131	0.9544	-10.9368	25.3396	-0.3396	0.1153

2. PEMULUSAN WINTER ADITIF

Procedures of Additive Holt-Winters Method

p -step-ahead forecast made at time T

$$\hat{y}_{T+p}(T) = \lambda_T + pb_T + sn_{T+p-L} \quad (p = 1, 2, 3, \dots)$$

Example

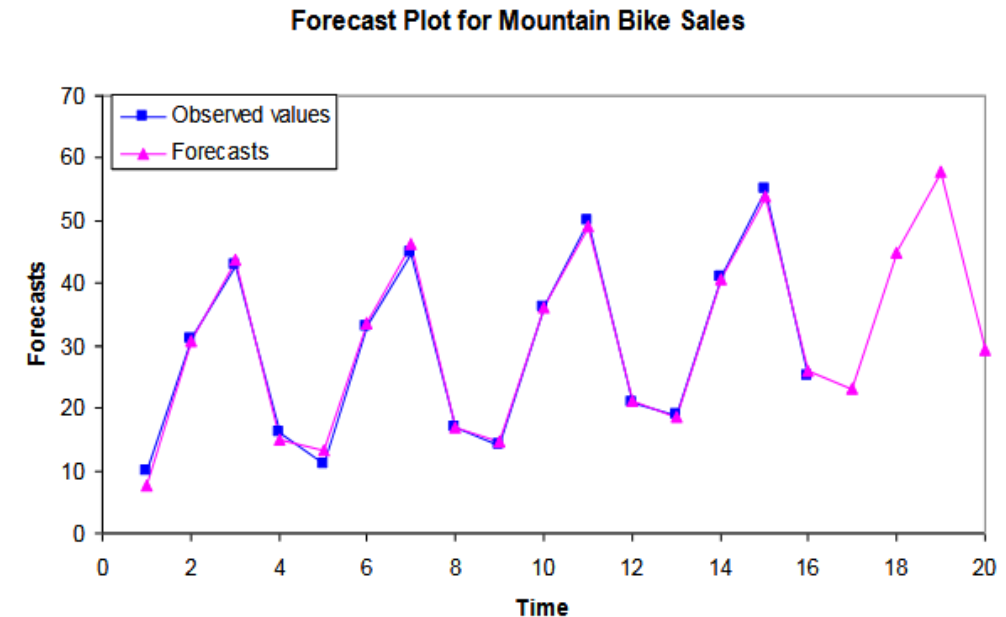
$$\hat{y}_{17}(16) = \lambda_{16} + b_{16} + sn_{17-4} = 36.3426 + 0.9809 - 14.2162 = 23.1073$$

$$\hat{y}_{18}(16) = \lambda_{16} + 2b_{16} + sn_{18-4} = 36.3426 + 2(0.9809) + 6.5529 = 44.8573$$

$$\hat{y}_{19}(16) = \lambda_{16} + 3b_{16} + sn_{19-4} = 36.3426 + 3(0.9809) + 18.5721 = 57.8573$$

$$\hat{y}_{20}(16) = \lambda_{16} + 4b_{16} + sn_{20-4} = 36.3426 + 4(0.9809) - 10.9088 = 29.3573$$

Example



OUTLINE

1. Pemulusan untuk Data Musiman

2. Pemulusan Winter Aditif

3. Pemulusan Winter Multiplikatif

3. PEMULUSAN WINTER MULTIPLIKATIF

Multiplicative Model

$$Y_t = (L_t)(S_t) + \varepsilon_t$$

level or linear trend component
can in turn be represented by $\beta_0 + \beta_1 t$

the seasonal adjustment

$$S_t = S_{t+m} = S_{t+2m} = \dots \text{ for } t = 1, \dots, m-1$$

length of the season (period) of the cycles

$$\sum_{t=1}^m S_t = 0.$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Holt-Winters Multiplicative Formulation

- Suppose the time series is denoted by $y_1, \dots, y_n$ with m seasonal period.

Estimate of the level:

$$L_t = \alpha \left(\frac{Y_t}{S_{t-m}} \right) + (1 - \alpha)(L_{t-1} + B_{t-1})$$

Estimate of the trend:

$$B_t = \gamma(L_t - L_{t-1}) + (1 - \gamma)B_{t-1}$$

Estimate of the seasonal factor:

$$S_t = \delta \left(\frac{Y_t}{L_t} \right) + (1 - \delta)S_{t-m}$$

- Let $\hat{Y}_{t+h}(t)$ be the h -step forecast made using data to time t

$$\hat{Y}_{t+h}(t) = (L_t + B_t h) S_{t+h-m}$$

3. PEMULUSAN WINTER MULTIPLIKATIF

The Procedure

Step 1: Initialize the value of L_t , B_t , and S_t

Step 2: Update the estimate of L_t

Step 3: Update the estimate of B_t

Step 4: Update the estimate of S_t

Step 5: Conduct the h -step-ahead forecast

3. PEMULUSAN WINTER MULTIPLIKATIF

Initializing the Holt-Winters method

Montgomery (2015):

Suppose a dataset consist of k seasons.

- $\hat{L}_0 = \frac{\bar{y}_k - \bar{y}_1}{(k-1)m}$ where $\bar{y}_i = \frac{1}{m} \sum_{t=(i-1)m+1}^{im} y_t$
- $\hat{B}_0 = \bar{y}_1 - \frac{m}{2} \hat{L}_0$
- $\hat{S}_{j-m} = m \left(\frac{\hat{S}_j^*}{\sum_{i=1}^m \hat{S}_j^*} \right)$, for $1 \leq j \leq s$, where $\hat{S}_j^* = \frac{1}{k} \sum_{t=1}^k \frac{y_{(t-1)m+j}}{\bar{y}_t - \left(\frac{s+1}{2-j} \right) \hat{\beta}_0}$

3. PEMULUSAN WINTER MULTIPLIKATIF

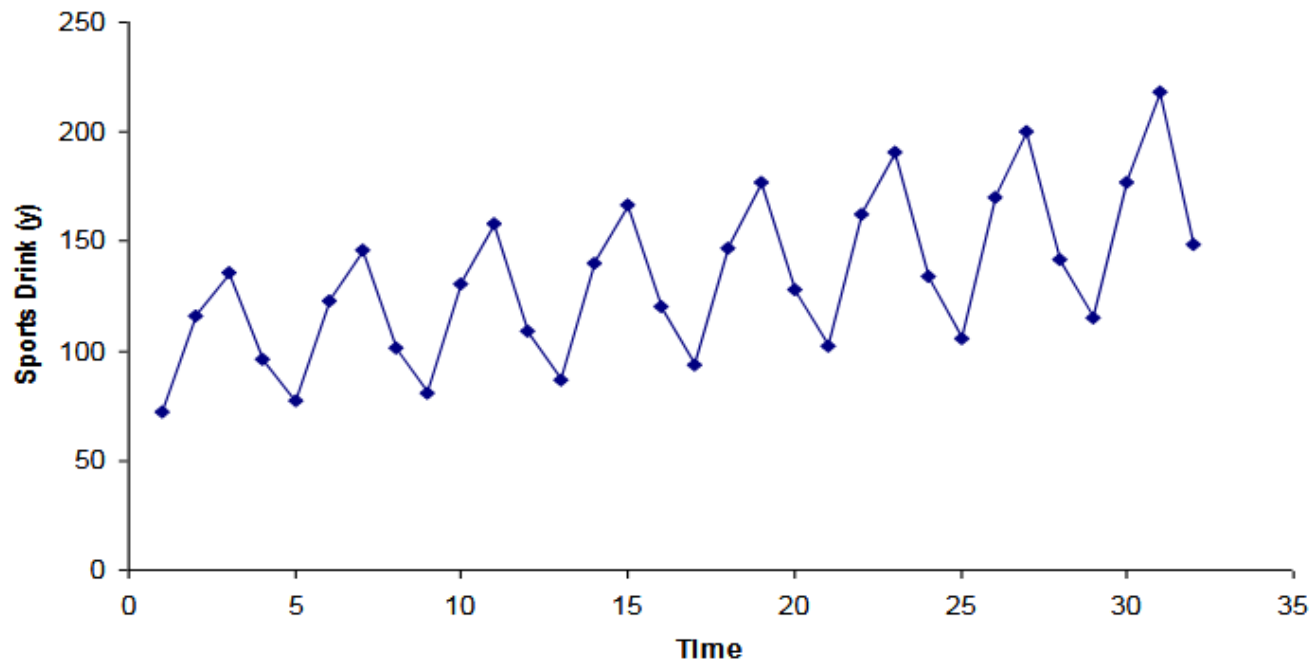
Procedures of Multiplicative Holt-Winters Method

Use the Sports Drink example as an illustration

Quarterly sales of Tiger Sports Drink								
	Year							
Quarter	1	2	3	4	5	6	7	8
1	72	77	81	87	94	102	106	115
2	116	123	131	140	147	162	170	177
3	136	146	158	167	177	191	200	218
4	96	101	109	120	128	134	142	149

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method



Observations:

- Linear upward trend over the 8-year period
- Magnitude of the seasonal span increases as the level of the time series increases

⇒ Multiplicative Holt-Winters method can be applied to forecast future sales

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

Step 1: Obtain initial values for the level ℓ_0 , the growth rate b_0 , and the seasonal factors s_{-3} , s_{-2} , s_{-1} , and s_0 , by fitting a least squares trend line to at least four or five years of the historical data.

- y-intercept = ℓ_0 ; slope = b_0

Example

- Fit a least squares trend line to the first 16 observations
- Trend line

$$\hat{y}_t = 95.2500 + 2.4706t$$

- $\ell_0 = 95.2500$; $b_0 = 2.4706$

SUMMARY OUTPUT	
<i>Regression Statistics</i>	
Multiple R	0.403809754
R Square	0.163062318
Adjusted R Square	0.103281055
Standard Error	27.58325823
Observations	16
ANOVA	
	<i>df</i>
Regression	1
Residual	14
Total	15
<i>Coefficients</i>	
Intercept	95.25
X Variable 1	2.470588235

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

Step 2: Find the initial seasonal factors

1. Compute $\hat{y}_t$ for the in-sample observations used for fitting the regression. In this example, $t = 1, 2, \dots, 16$.

$$\hat{y}_1 = 95.2500 + 2.4706(1) = 97.7206$$

$$\hat{y}_2 = 95.2500 + 2.4706(2) = 100.1912$$

.....

$$\hat{y}_{16} = 95.2500 + 2.4706(16) = 134.7794$$

Step 2: Find the initial seasonal factors

2. Detrend the data by computing $S_{0,t} = y_t / \hat{y}_t$ for each time period that is used in finding the least squares regression equation. In this example, $t = 1, 2, \dots, 16$.

$$S_{0,1} = y_1 / \hat{y}_1 = 72 / 97.7206 = 0.7368$$

$$S_{0,2} = y_2 / \hat{y}_2 = 116 / 100.1912 = 1.1578$$

..... ..

$$S_{0,16} = y_{16} / \hat{y}_{16} = 120 / 134.7794 = 0.8903$$

Step 2: Find the initial seasonal factors

3. Compute the average seasonal values for each of the k seasons. The k averages are found by computing the average of the detrended values for the corresponding season. For example, for quarter 1,

$$\begin{aligned}\bar{S}_{[1]} &= \frac{S_1 + S_5 + S_9 + S_{13}}{4} \\ &= \frac{0.7368 + 0.7156 + 0.6894 + 0.6831}{4} = 0.7062\end{aligned}$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

Step 2: Find the initial seasonal factors

4. Multiply the average seasonal values by the normalizing constant

$$CF = \frac{m}{\sum_{i=1}^m \bar{S}_{[i]}}$$

such that the average of the seasonal factors is 1. The initial seasonal factors are

$$S_{i-m} = \bar{S}_{[i]}(CF) \quad (i = 1, 2, \dots, L)$$

Step 2: Find the initial seasonal factors

4. Multiply the average seasonal values by the normalizing constant such that the average of the seasonal factors is 1.

- Example

$$CF = 4/3.9999 = 1.0000$$

$$S_{-3} = S_{1-4} = \bar{S}_{[1]}(CF) = 0.7062(1) = 0.7062$$

$$S_{-2} = S_{2-4} = \bar{S}_{[2]}(CF) = 1.1114(1) = 1.1114$$

$$S_{-1} = S_{3-4} = \bar{S}_{[3]}(CF) = 1.2937(1) = 1.2937$$

$$S_0 = S_{4-4} = \bar{S}_{[4]}(CF) = 0.8886(1) = 0.8886$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

Step 3: Calculate a point forecast of y_1 from time 0 using the initial values

$$\hat{y}_{t+h}(t) = (L_t + B_t h) S_{t+h-m} \quad (t = 1, h = 0)$$

$$\begin{aligned}\hat{y}_1(0) &= (L_0 + B_0) S_{1-4} = (L_0 + B_0) S_{-3} \\ &= (95.25 + 2.4706) 0.7062 \\ &= 69.0103\end{aligned}$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

Step 4: Update the estimates ℓ_T , b_T , and sn_T by using some predetermined values of smoothing constants.

Example: let $\alpha = 0.2$, $\gamma = 0.1$, and $\delta = 0.1$

$$\begin{aligned}l_1 &= \alpha(y_1 / s_{1-4}) + (1 - \alpha)(l_0 + b_0) \\ &= 0.2(72 / 0.7062) + 0.8(95.2500 + 2.4706) = 98.5673\end{aligned}$$

$$\begin{aligned}b_1 &= \gamma(l_1 - l_0) + (1 - \gamma)b_0 \\ &= 0.1(98.5673 - 95.2500) + 0.9(2.4706) = 2.5553\end{aligned}$$

$$\begin{aligned}sn_1 &= \delta(y_1 / l_1) + (1 - \delta)s_{1-4} \\ &= 0.1(72 / 98.5673) + 0.9(0.7062) = 0.7086\end{aligned}$$

$$\begin{aligned}\hat{y}_2(1) &= (l_1 + b_1)s_{2-4} \\ &= (98.5673 + 2.5553)(1.1114) = 112.3876\end{aligned}$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

$$\begin{aligned}\lambda_2 &= \alpha(y_2 / s_{2-4}) + (1 - \alpha)(\lambda_1 + b_1) \\ &= 0.2(116/1.1114) + 0.8(98.5673 + 2.5553) \\ &= 101.7727\end{aligned}$$

$$\begin{aligned}b_2 &= \gamma(\lambda_2 - \lambda_1) + (1 - \gamma)b_1 \\ &= 0.1(101.7727 - 98.5673) + 0.9(2.5553) \\ &= 2.62031\end{aligned}$$

$$\begin{aligned}sn_2 &= \delta(y_2 / \lambda_2) + (1 - \delta)s_{2-4} \\ &= 0.1(116/101.7727) + 0.9(1.1114) \\ &= 1.114239\end{aligned}$$

$$\begin{aligned}\hat{y}_3(2) &= (\lambda_2 + b_2)s_{3-4} \\ &= (101.7727 + 2.62031)(1.2937) \\ &= 135.053\end{aligned}$$

$$\begin{aligned}\lambda_4 &= \alpha(y_4 / s_{4-4}) + (1 - \alpha)(\lambda_3 + b_3) \\ &= 0.2(96/0.8886) + 0.8(104.5393 + 2.6349) \\ &= 107.3464\end{aligned}$$

$$\begin{aligned}b_4 &= \gamma(\lambda_4 - \lambda_3) + (1 - \gamma)b_3 \\ &= 0.1(107.3464 - 104.5393) + 0.9(2.6349) \\ &= 2.65212\end{aligned}$$

$$\begin{aligned}sn_4 &= \delta(y_4 / \lambda_4) + (1 - \delta)s_{4-4} \\ &= 0.1(96/107.3464) + 0.9(0.8886) \\ &= 0.889170\end{aligned}$$

$$\begin{aligned}\hat{y}_5(4) &= (\lambda_4 + b_4)s_{5-4} \\ &= (107.3464 + 2.65212)(0.7086) \\ &= 77.945\end{aligned}$$

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

1	n	alpha	gamma	delta	SSE	MSE	s	
2	32	0.2	0.1	0.1	177.3223	6.1146	2.4728	
3								
4								
5						Forecast		Squared
6				Growth	Seasonal	Made Last	Forecast	Forecast
7	Time	y	Level	Rate	Factor	Period	Error	Error
8	-3				0.7062			
9	-2				1.1114			
10	-1				1.2937			
11	0		95.25	2.4706	0.8886			
12	1	72	98.56729	2.5553	0.7086	69.0103	2.9897	8.9384
13	2	116	101.7726	2.6203	1.1142	112.3876	3.6124	13.0494
14	3	136	104.5393	2.6349	1.2944	135.0531	0.9469	0.8967
15	4	96	107.3464	2.6521	0.8892	95.2350	0.7650	0.5853
16	5	77	109.731	2.6254	0.7079	77.9478	-0.9478	0.8984
17	6	123	111.9629	2.5860	1.1127	125.1919	-2.1919	4.8043
18	7	146	114.1974	2.5509	1.2928	148.2750	-2.2750	5.1755
19	8	101	116.1165	2.4877	0.8872	103.8091	-2.8091	7.8911
20	9	81	117.7668	2.4040	0.7059	83.9641	-2.9641	8.7858
21	10	131	119.6835	2.3552	1.1109	133.7108	-2.7108	7.3482
22	11	158	122.0734	2.3587	1.2930	157.7754	0.2246	0.0504
23	12	109	124.1164	2.3271	0.8863	110.4005	-1.4005	1.9615
24	13	87	125.8035	2.2631	0.7045	89.2593	-2.2593	5.1044
25	14	140	127.6589	2.2224	1.1094	142.2642	-2.2642	5.1268
26	15	167	129.7369	2.2079	1.2924	167.9337	-0.9337	0.8718

.....								
38	27	200	156.1396	2.1752	1.2903	202.0396	-2.0396	4.1601
39	28	142	158.5505	2.1988	0.8908	140.9508	1.0492	1.1008
40	29	115	161.2803	2.2519	0.7047	113.1314	1.8686	3.4918
41	30	177	162.8178	2.1804	1.1046	180.9529	-3.9529	15.6252
42	31	218	165.7889	2.2595	1.2928	212.8988	5.1012	26.0220
43	32	149	167.8899	2.2437	0.8905	149.7057	-0.7057	0.4981

3. PEMULUSAN WINTER MULTIPLIKATIF

Procedures of Multiplicative Holt-Winters Method

p -step-ahead forecast made at time T

$$\hat{y}_{t+h}(t) = (L_t + B_t h) S_{t+h-m} \quad (h = 1, 2, 3, \dots)$$

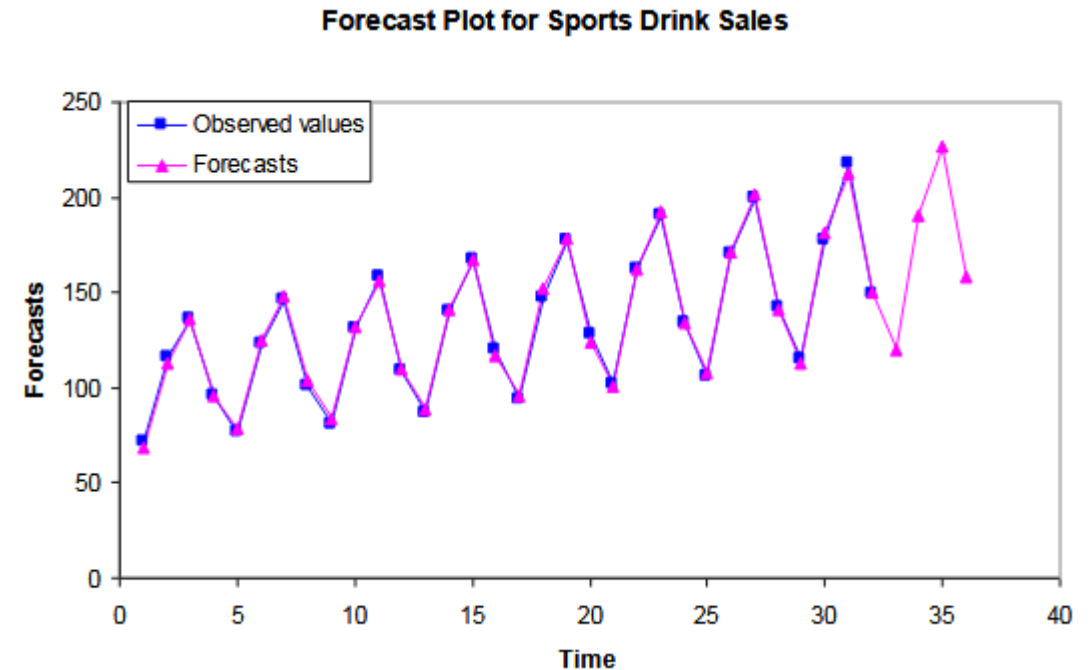
Example

$$\hat{y}_{33}(32) = (l_{32} + b_{32}) s_{33-4} = (168.1213 + 2.3028)(0.7044) = 120.0467$$

$$\hat{y}_{34}(32) = (l_{32} + 2b_{32}) s_{34-4} = [168.1213 + 2(2.3028)](1.1038) = 190.6560$$

$$\hat{y}_{35}(32) = (l_{32} + 3b_{32}) s_{35-4} = [(168.1213 + 3(2.3028)](1.2934) = 226.3834$$

$$\hat{y}_{36}(32) = (l_{32} + 4b_{32}) s_{36-4} = [(168.1213 + 4(2.3028)](0.8908) = 157.9678$$



3. PEMULUSAN WINTER MULTIPLIKATIF

Holt-Winters Additive Vs Multiplicative Formulation

Suppose the time series is denoted by $y_1, \dots, y_n$ with m seasonal period.

	Additive	Multiplicative
Est. of level	$L_t = \alpha(Y_t - S_{t-m}) + (1 - \alpha)(L_{t-1} + B_{t-1})$	$L_t = \alpha\left(\frac{Y_t}{S_{t-m}}\right) + (1 - \alpha)(L_{t-1} + B_{t-1})$
Est. of trend	$B_t = \gamma(L_t - L_{t-1}) + (1 - \gamma)B_{t-1}$	$B_t = \gamma(L_t - L_{t-1}) + (1 - \gamma)B_{t-1}$
Est. of seasonal	$S_t = \delta(Y_t - L_t) + (1 - \delta)S_{t-m}$	$S_t = \delta\left(\frac{Y_t}{L_t}\right) + (1 - \delta)S_{t-m}$
Forecast	$\hat{Y}_{t+h}(t) = L_t + B_t h + S_{t+h-m}$	$\hat{Y}_{t+h}(t) = (L_t + B_t h)S_{t+h-m}$

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